



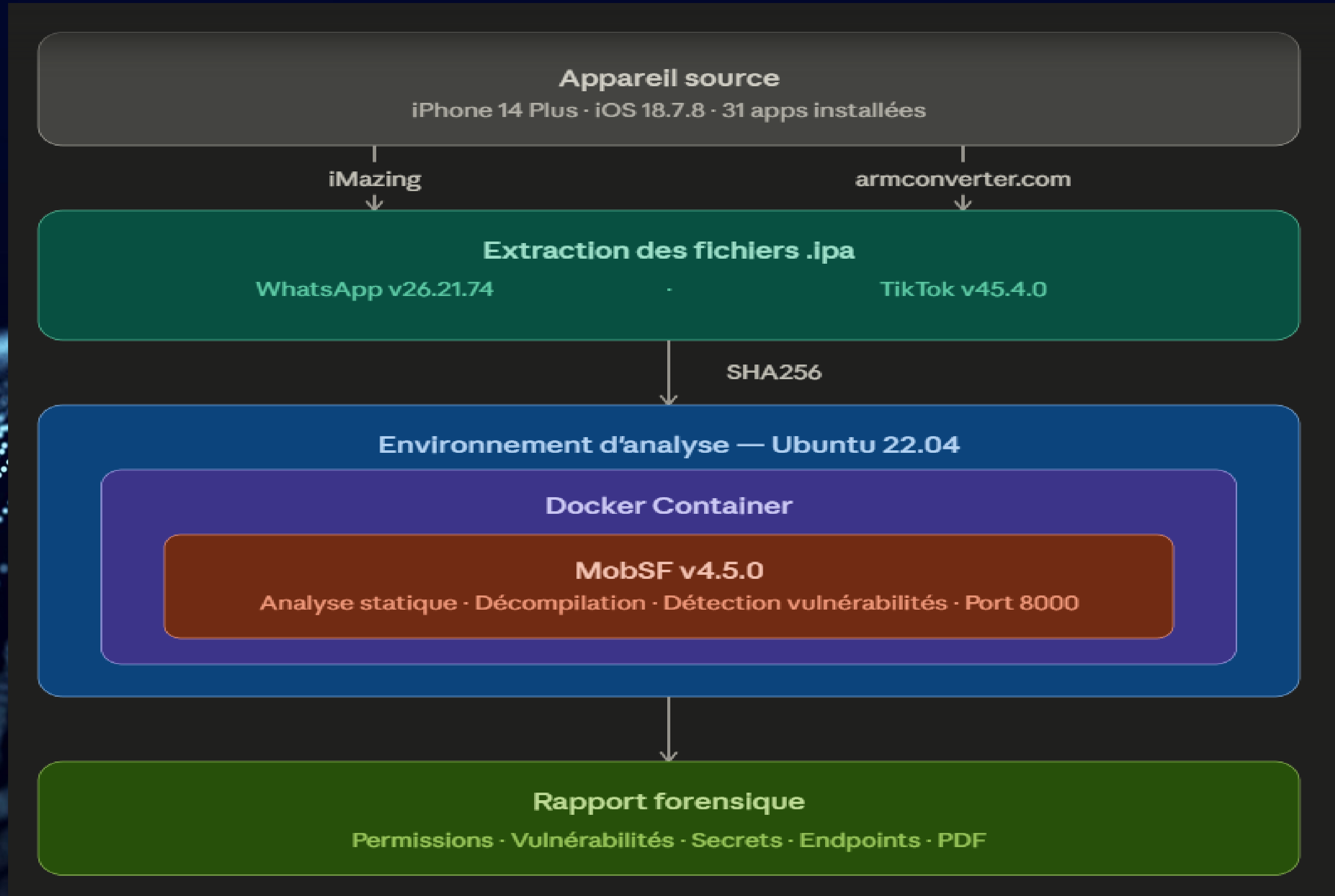
BEM.TECH

Réalisé par HAYE FALL



analyse forensic
avec mobsf





sommaire

Introduction à l'analyse forensique mobile

Présentation des outils utilisés

1. Environnement de travail

2. Méthodologie

3. Extraction des fichiers .ipa

4. Analyse statique – WhatsApp

5. Analyse statique – TikTok

6. Résultats et vulnérabilités détectées

Conclusion et recommandations

les outils utiliser

Outil	Usage	Installation
Docker	Faire tourner MobSF	docker.com
MobSF	Analyse principale	Via Docker
Python 3.8+	Scripts d'automatisation	python.org

MobSF (Mobile Security Framework) Cadre de travail open-source dédié à l'analyse de sécurité des applications mobiles. Il permet d'effectuer des analyses statiques et dynamiques sur les applications iOS et Android de manière automatisée.

iMazing Logiciel de gestion d'appareils Apple sous Windows permettant d'accéder aux applications installées sur un iPhone et d'extraire des informations forensiques.

Docker Plateforme de conteneurisation qui permet d'isoler et déployer des applications dans un environnement contrôlé et reproductible, sans affecter le système hôte.

Armconverter — Decrypted IPA Store Plateforme en ligne permettant de télécharger des fichiers .ipa décryptés des applications iOS, prêts pour l'analyse forensique.

SHA256 Algorithme de hachage cryptographique utilisé pour calculer l'empreinte numérique des fichiers et garantir leur intégrité tout au long de la chaîne de custody.

3 — Environnement de travail

Machine d'analyse

- Système d'exploitation : Ubuntu 22.04 LTS

Appareil analysé

- Modèle : iPhone 14 Plus
 - Système : iOS 18.7.8
- Opérateur : Orange Sénégal
 - Stockage : 128 GB

Applications analysées

Application	Version	Taille	Bundle ID
WhatsApp	26.21.74	155 MB	net.whatsapp.WhatsApp
TikTok	45.4.0	399 MB	com.zhiliaoapp.musically



4 – Méthodologie avec commandes

Phase 1 – Préparation de l'environnement

```
sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
```

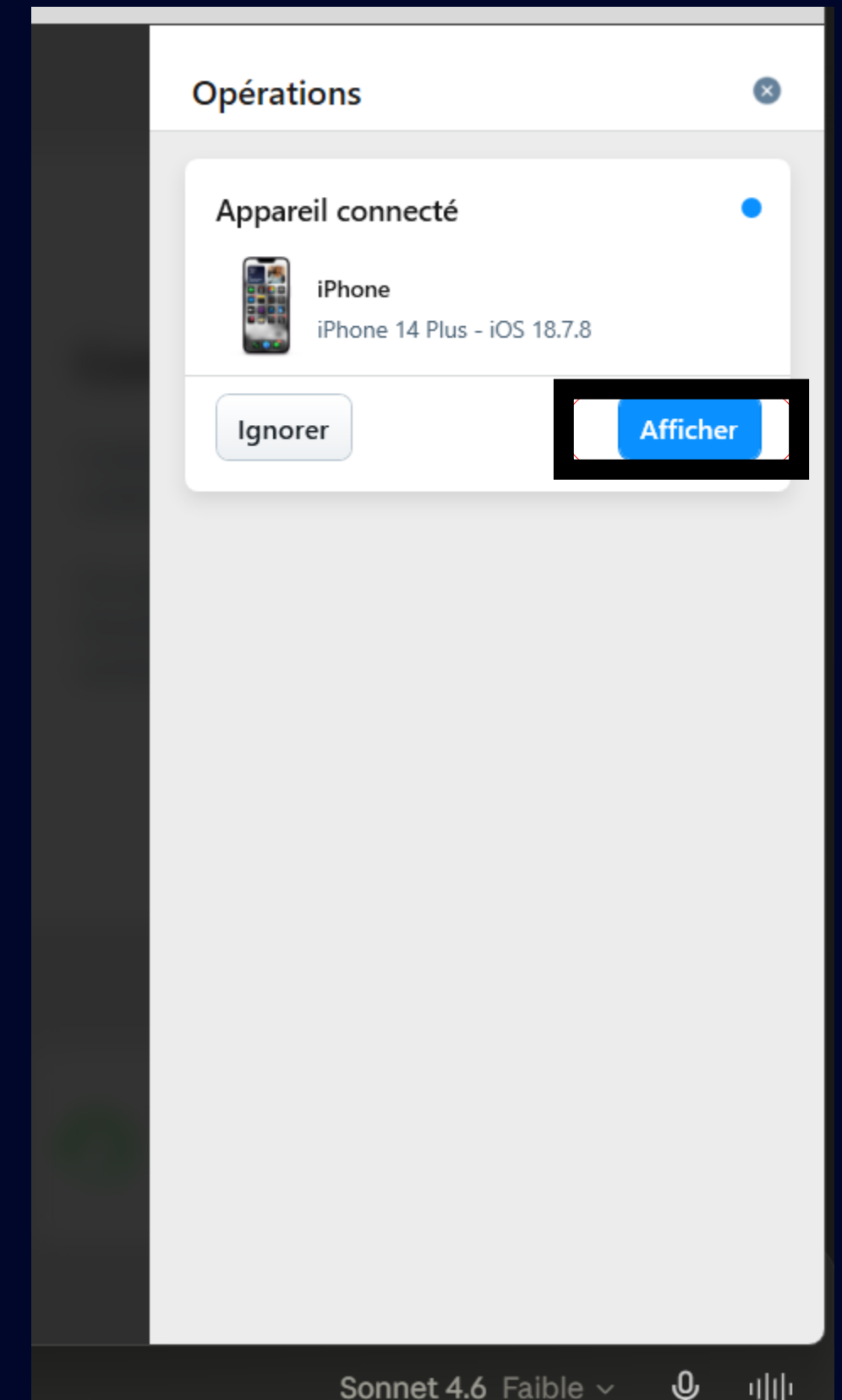
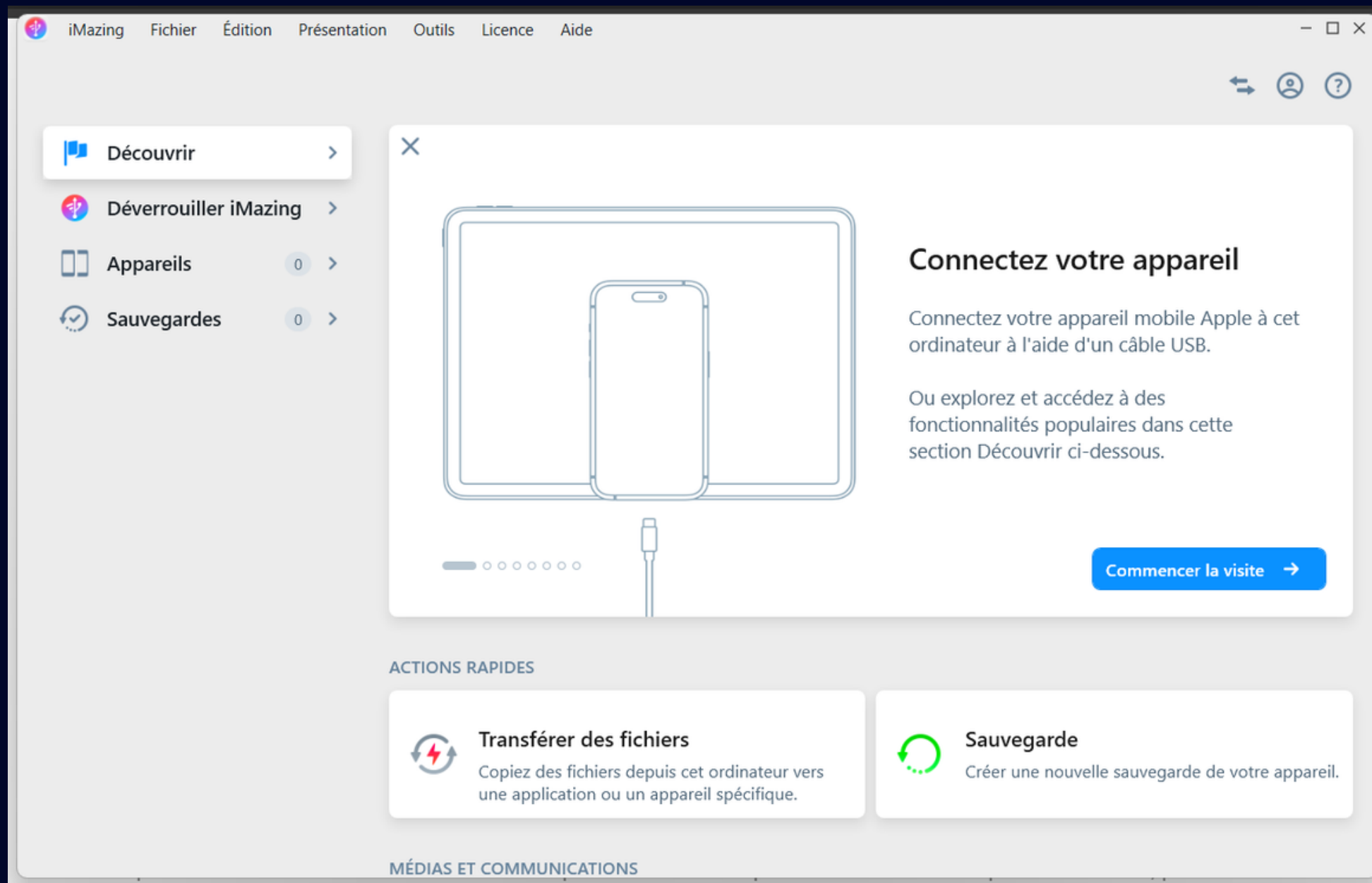
```
sudo docker run -it --rm -p 8000:8000 \  
--name mobsf \
```

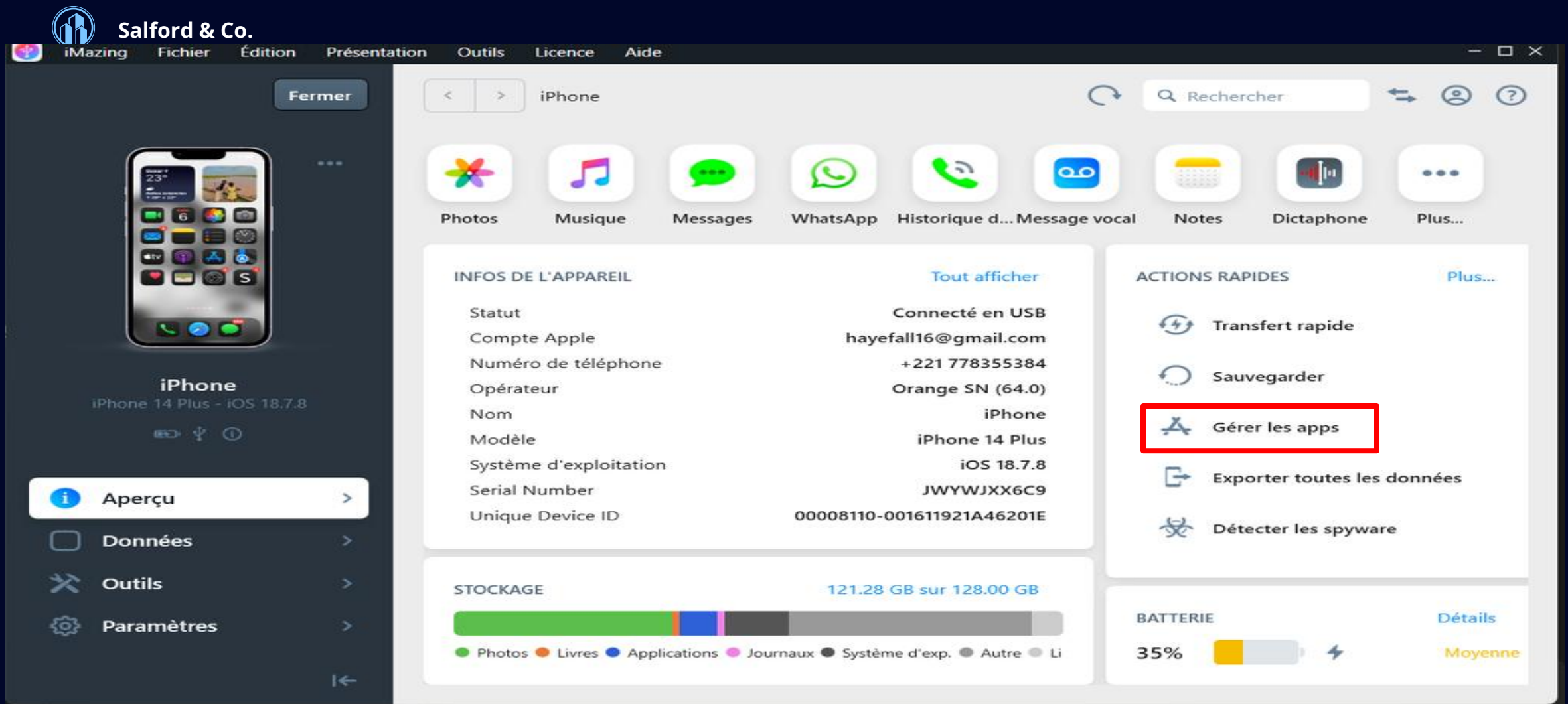
```
opensecurity/mobile-security-framework-mobsf:latest
```

```
mkdir -p ~/forensic_iphone
```

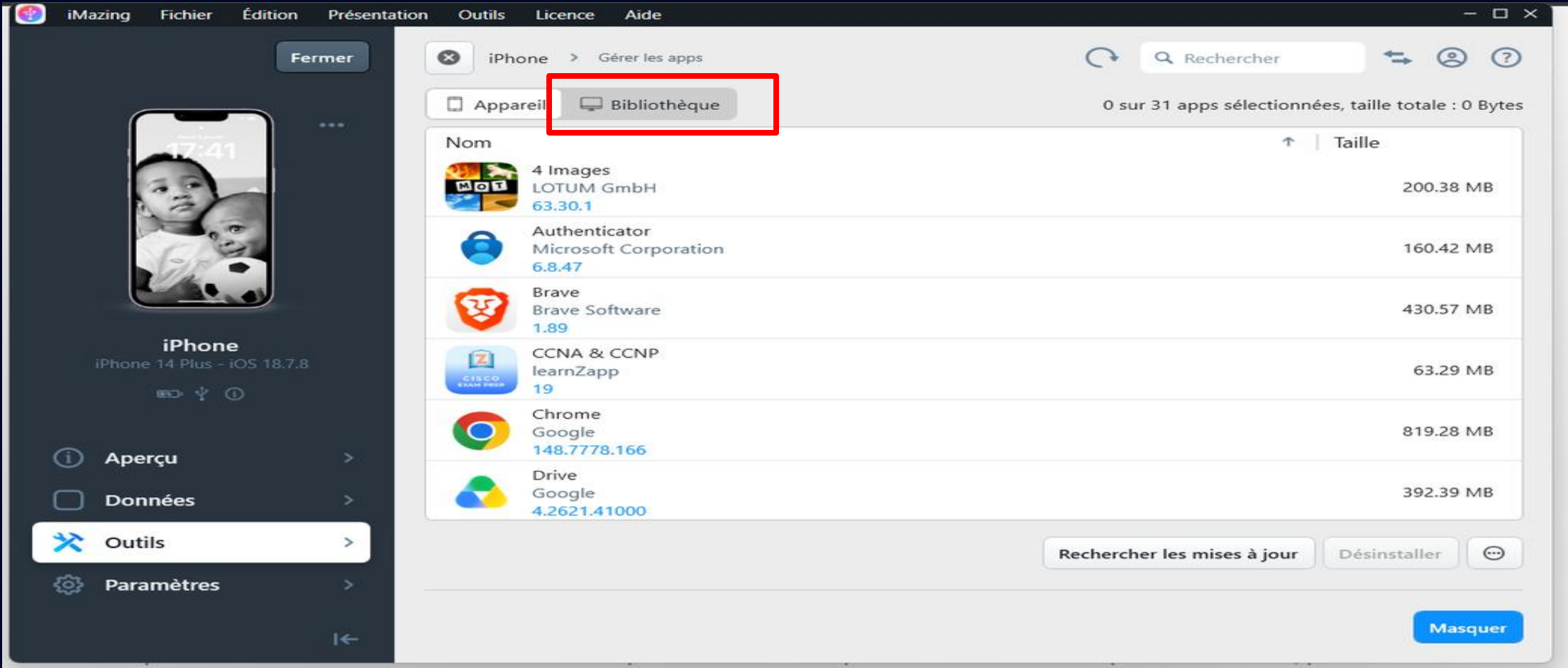
```
kane@kane-virtual-machine:~$ sudo docker pull opensecurity/mobile-security-frame  
work-mobsf:latest  
latest: Pulling from opensecurity/mobile-security-framework-mobsf  
f6eeee67085b: Downloading 4.194MB/105.1MB  
792aba8f9d95: Download complete  
8b6abb057bb6: Downloading 11.53MB/254.6MB  
219a2e93422d: Download complete  
6db0909c4473: Extracting 6 s  
0b03c5abb644: Download complete  
63706ab84aa5: Download complete  
19851ebfeaad: Download complete  
5f4836bfc719: Downloading 19.92MB/387.2MB  
aa0ff2f729c8: Downloading 17.83MB/105.1MB  
ae14b11b20fb: Downloading 7.34MB/96.25MB  
231aa360dc09: Download complete  
28ec1fde30db: Download complete  
3b91a9449057: Download complete  
unexpected EOF  
kane@kane-virtual-machine:~$
```

- Connexion iPhone via câble USB
- Ouverture de iMazing sur Windows
- Clic sur "Gérer les apps" → liste des 31 apps détectées





Une fois le téléphone brancher aller sur GERER
LES APPS puis aller sur BIBLIOTHEQUE

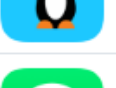
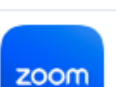


Appareil

Bibliothèque

@ hayefall16@gmail.com

Déconnexion

Nom	↑	Installé	Taille	Télécharge..
 Snapchat Snap, Inc. 13.94.0		✓	348.54 MB	↓
 TikTok TikTok Ltd. 45.2.0		✓	814.60 MB	↓
 To Do List BetterApp Tech Co., Limited 3.7.9		✓	129.81 MB	↓
 Wave Wave Mobile Money 26.5.28		✓	46.83 MB	↓
 WhatsApp WhatsApp Inc. 26.20.73		✓	305.45 MB	↓
 Yango YHub ZAF (Pty) Ltd 700.143.0		✓	473.39 MB	↓
 YouTube Google 21.22.4		✓	314.57 MB	↓
 Zoom Zoom Communications, Inc. 7.0.5		✓	303.40 MB	↓
 Zoom Rooms Zoom Communications, Inc. 7.0.6		✓	321.42 MB	↓

iPhone

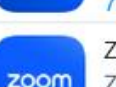
Gérer les apps

Rechercher

1 sur 31 apps sélectionnées, taille totale : 305.45 MB

Appareil

Bibliothèque

Nom	↑	Taille
 Wave Wave Mobile Money 26.5.28		71.79 MB
 WhatsApp WhatsApp Inc. 26.20.73		305.45 MB
 Yango YHub ZAF (Pty) Ltd 700.143.0		480.66 MB
 YouTube Google 21.22.4		2.92 GB
 Zoom Zoom Communications, Inc. 7.0.5		369.83 MB
 Zoom Rooms Zoom Communications, Inc. 7.0.6		321.42 MB

Désinstaller

Sauvegarder données appli

Restaurer données appli

Installer le fichier .IPA

Rechercher les mises à jour

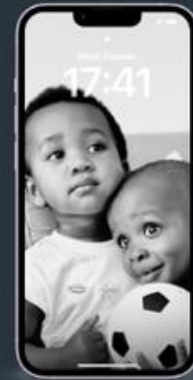
Exporter la liste en CSV

Rechercher les mises à jour

Désinstaller

Masquer

Fermer



iPhone

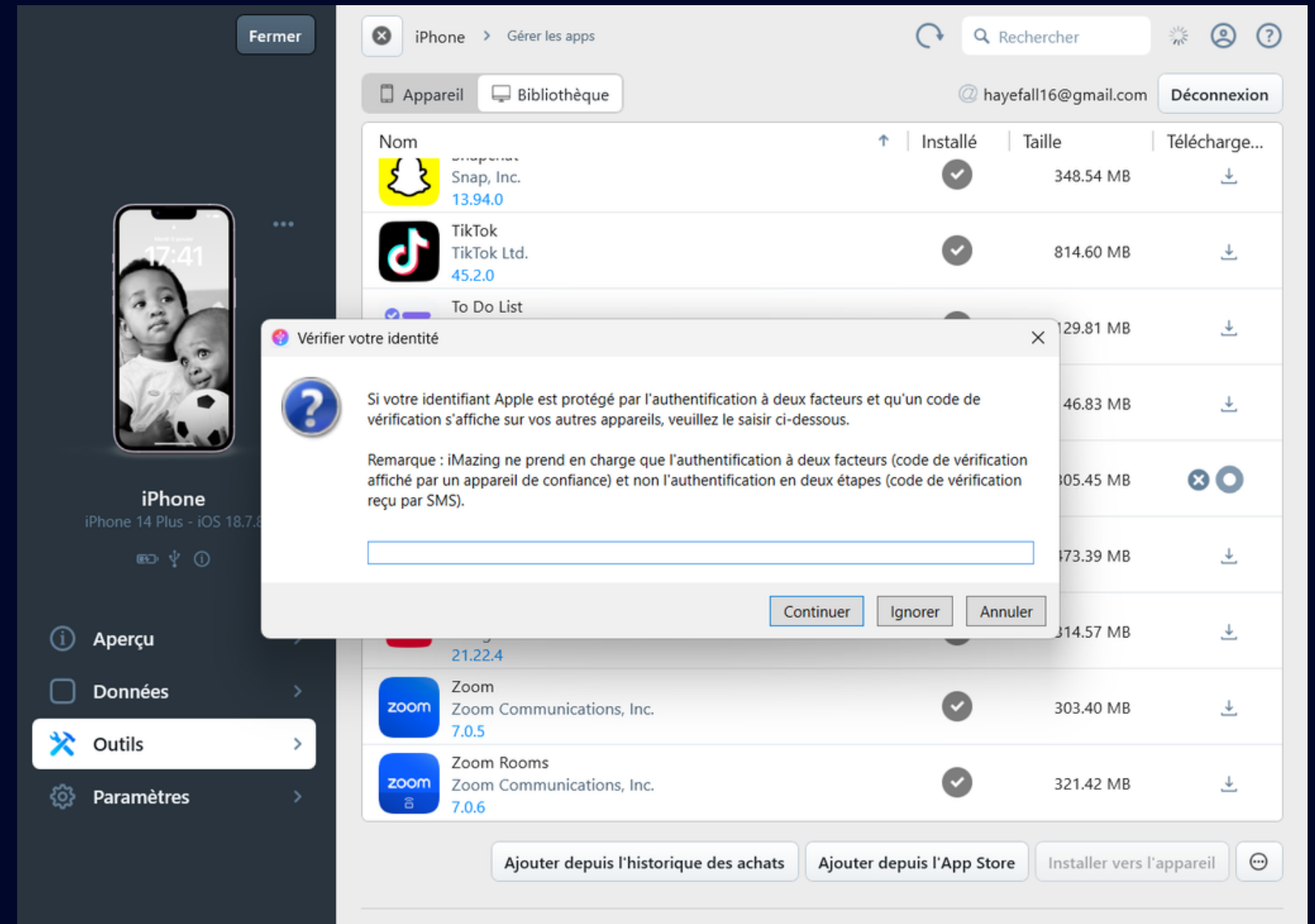
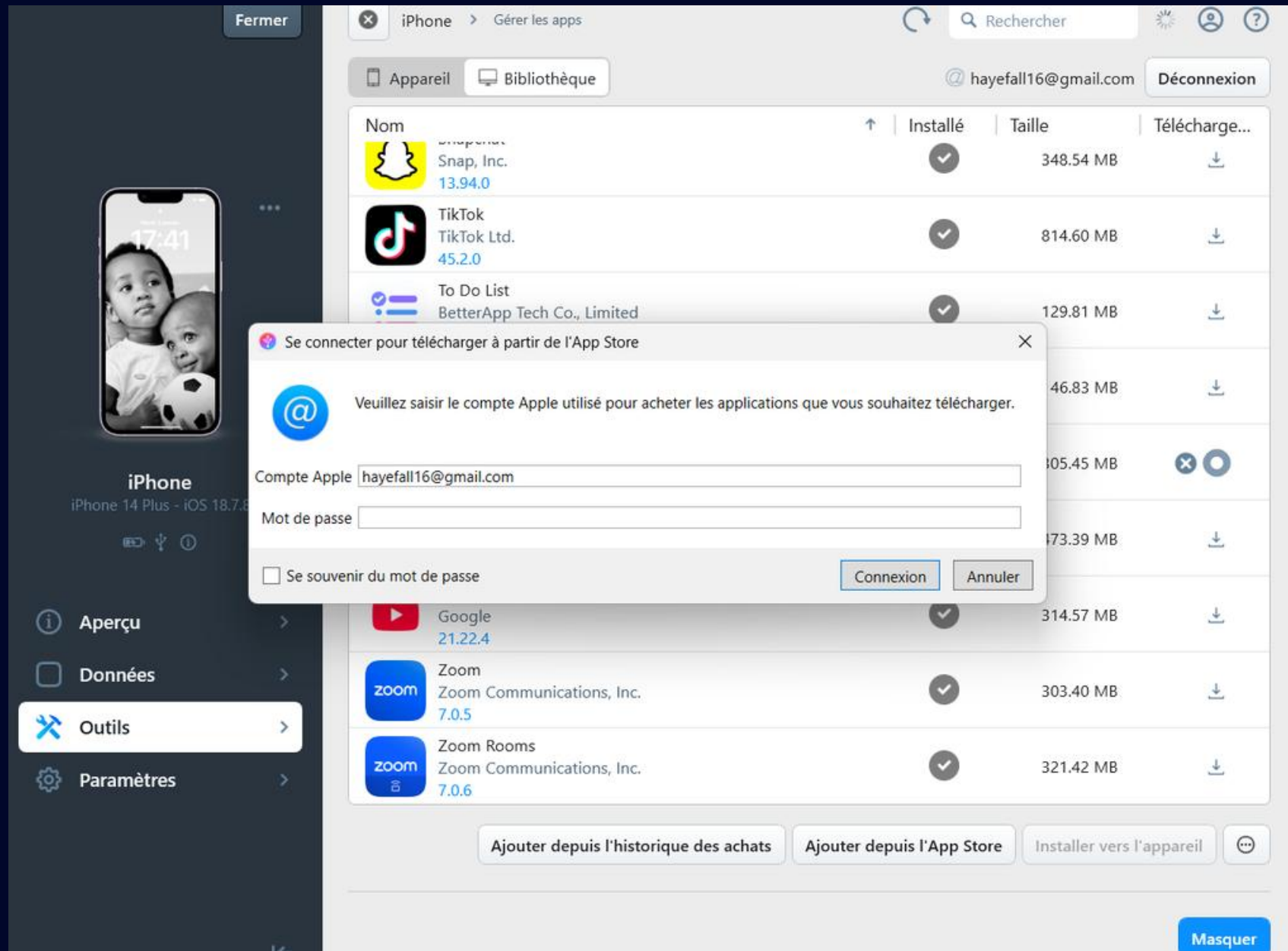
iPhone 14 Plus - iOS 18.7.8

Aperçu

Données

Outils

Paramètres





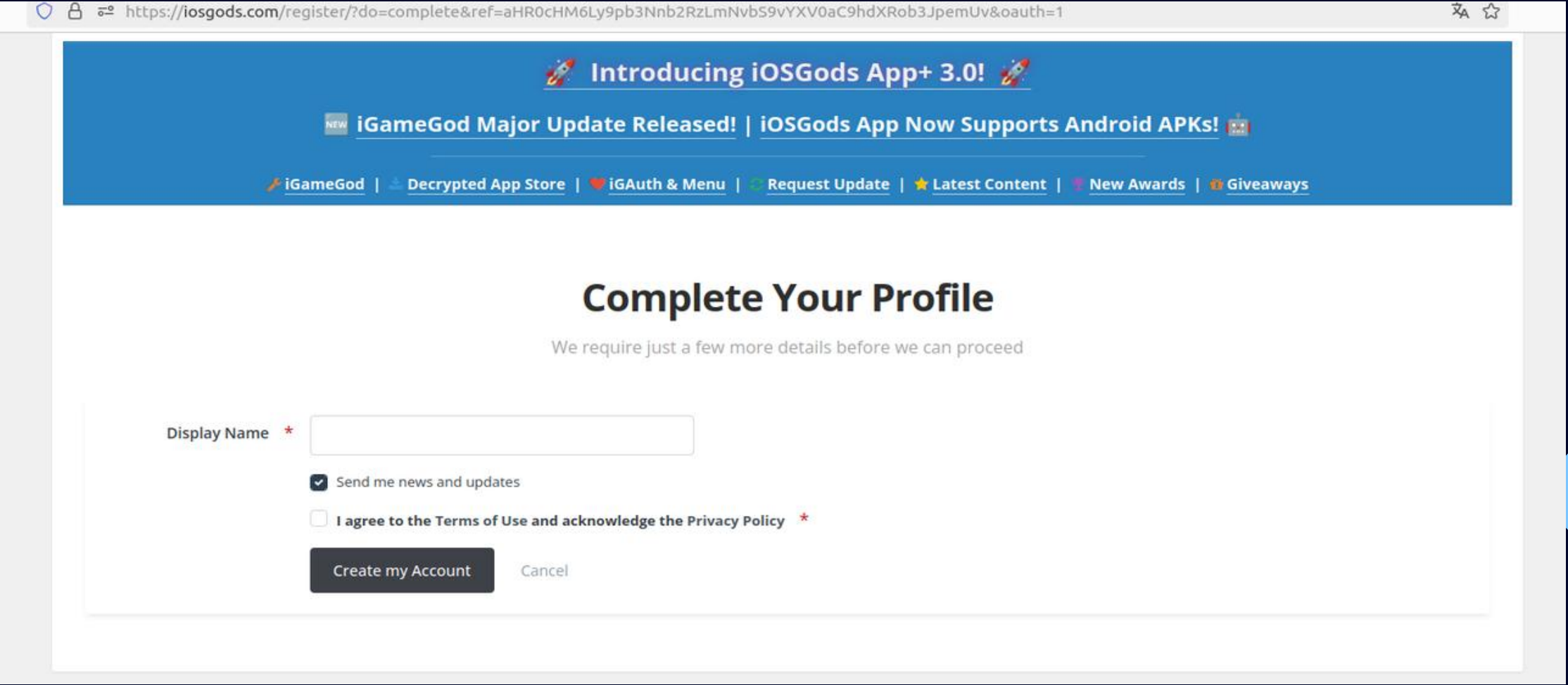
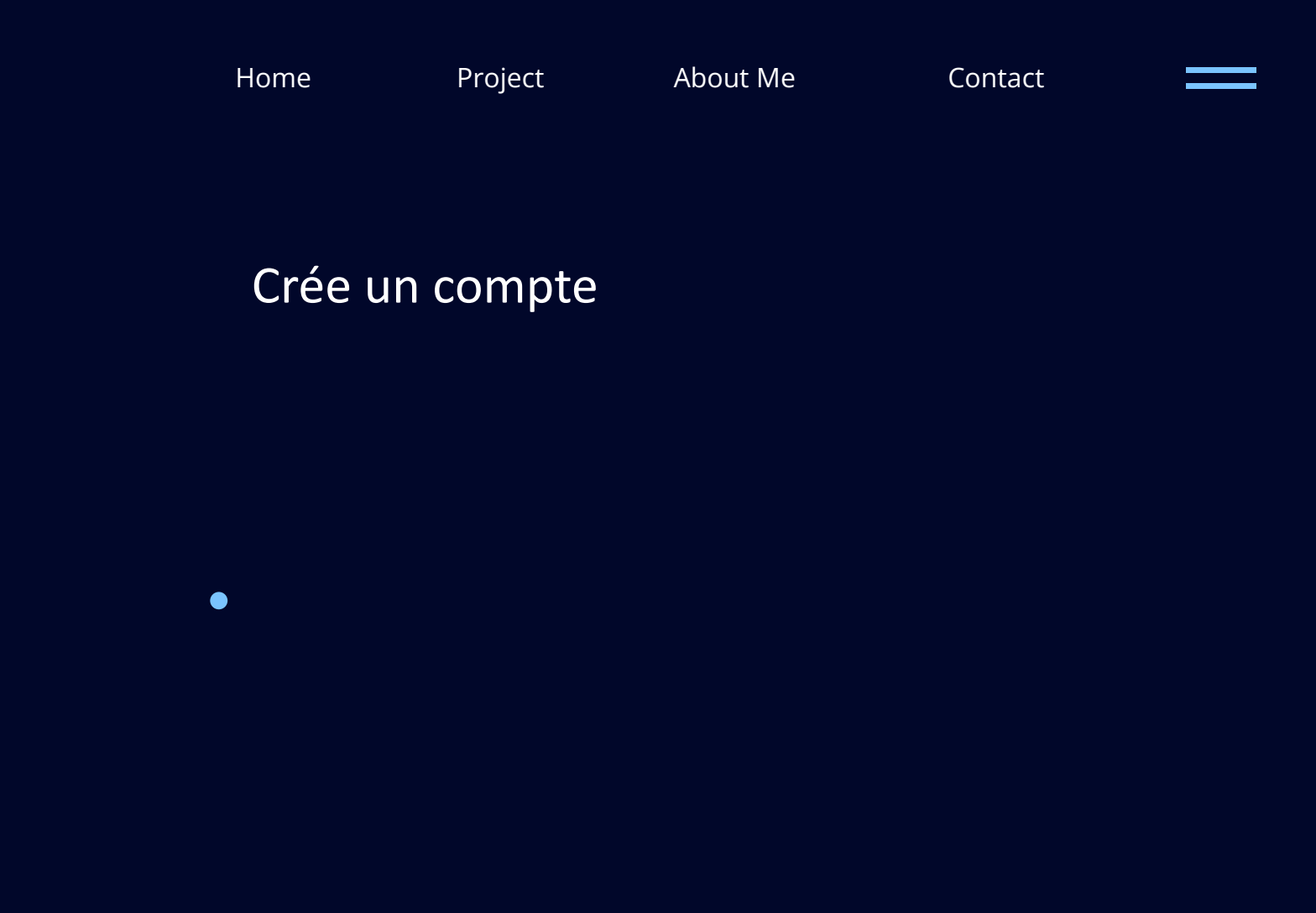
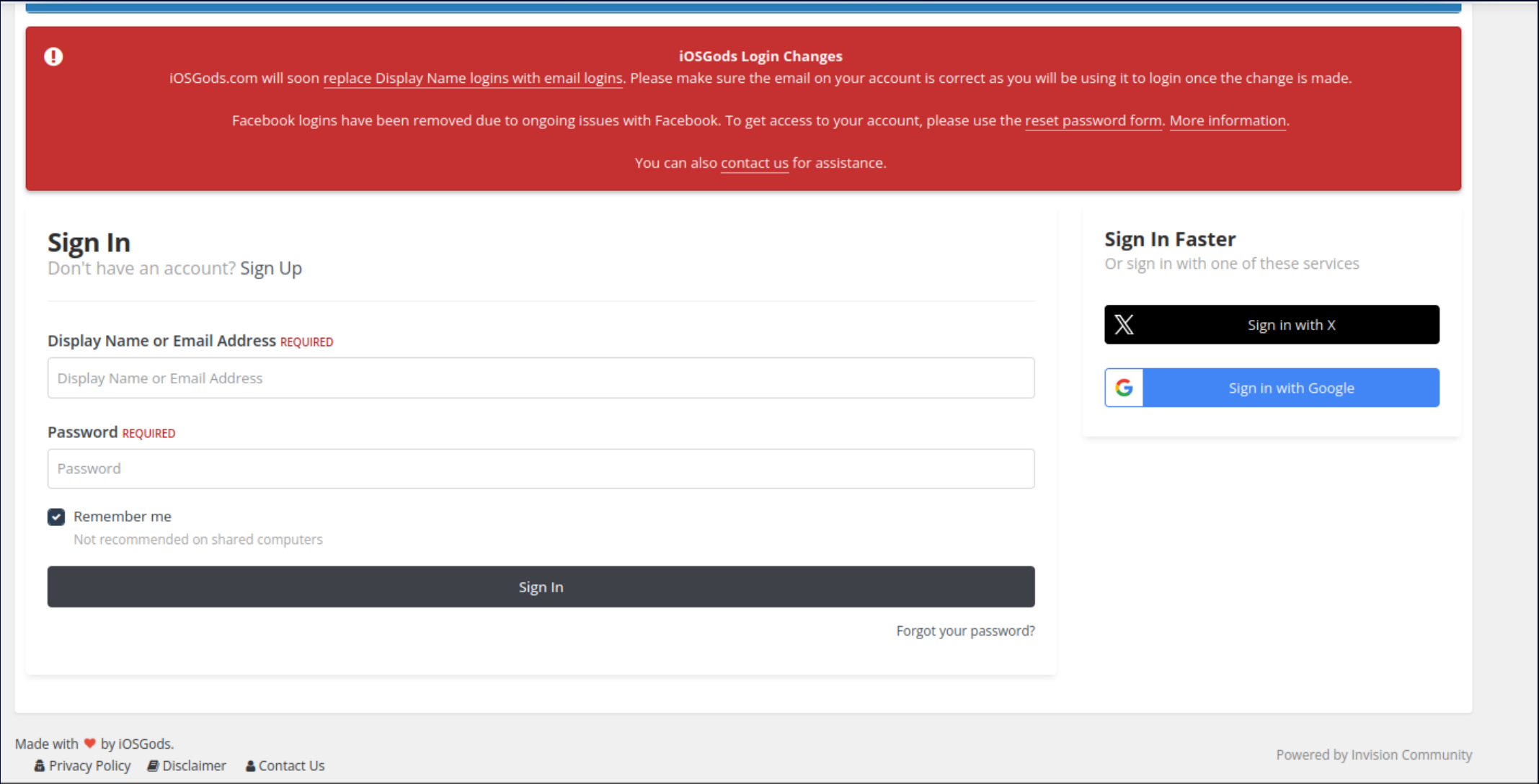
3 — Extraction des fichiers

.ipa

```
# Téléchargement depuis armconverter.com  
# Déplacement dans le dossier de travail  
mv ~/Téléchargements/*.ipa ~/forensic_iphone/
```

```
# Vérification intégrité SHA256  
cd ~/forensic_iphone  
sha256sum *.ipa > hashes.txt  
cat hashes.txt
```

une fois accédé au site il faudra crée un compte





Computer X SELKS1 X cible X cible1 X SELKS X kali X ubuntu serveur X kali1 X nextcloud X

tés Navigateur Web Firefox 7 juin 00:31

Restauration de session X [Tool] Online Decrypted i X Nouvel onglet X Decrypted iOS IPA App S X World of Warships: des b X World of Warships: des b X

https://armconverter.com/decryptedappstore/us/whatsapp

Decrypted iOS IPA App Store

Home > Decrypted IPA Store > Search Results

whatsapp

Country USA SEARCH

01 WhatsApp Messenger

02 by WhatsApp Inc. v26.21.74

Bundle ID: net.whatsapp.WhatsApp

Compatibility: iOS 15.1+

Size: 367,6 MB

Download state: ready

03 DOWNLOAD IPA

INSTALL WITH SIDELOADLY

Instagram

by Instagram, Inc. v432.0.0

Bundle ID: com.burbn.instagram

Compatibility: iOS 16.3+

Size: 575,9 MB

DECRYPT THIS APP

Messenger

by Meta Platforms, Inc. v563.0.0

Bundle ID: com.facebook.Messenger

Compatibility: iOS 15.1+

Size: 237,8 MB

DECRYPT THIS APP

YouTube

by Google v21.22.04

Bundle ID: com.google.ios.youtube

Compatibility: iOS 16.0+

Size: 364,9 MB

DECRYPT THIS APP

net.whatsapp.WhatsApp-26.21.74-Decrypted.ipa

Temps restant : 3 min 18 s — 6,5 sur 155 Mo (696 Ko/s)

Afficher tous les téléchargements

et on fait la même chose avec l'application tiktok pour télécharger le fichier .IPA



verification de l'intégrité des fichier .IPA

```
kane@kane-virtual-machine:~$ cd ~/forensic_iphone
sha256sum *.ipa > hashes.txt
cat hashes.txt
fc6db9b0ea67d65d1fd3bc95c2c952e02b8dd4ef750d26e2811597cacd3d69e2  com.zhiliaoapp
.musically-45.4.0-Decrypted.ipa
ee88f31441dcc10bb4fc205fb587bf83d872afc7f762444d67865a5031d96ed0  net.whatsapp.W
hatsApp-26.21.74-Decrypted.ipa
kane@kane-virtual-machine:~/forensic_iphone$
```

4 — Analyse statique

Vérifier que MobSF tourne
`curl http://localhost:8000`

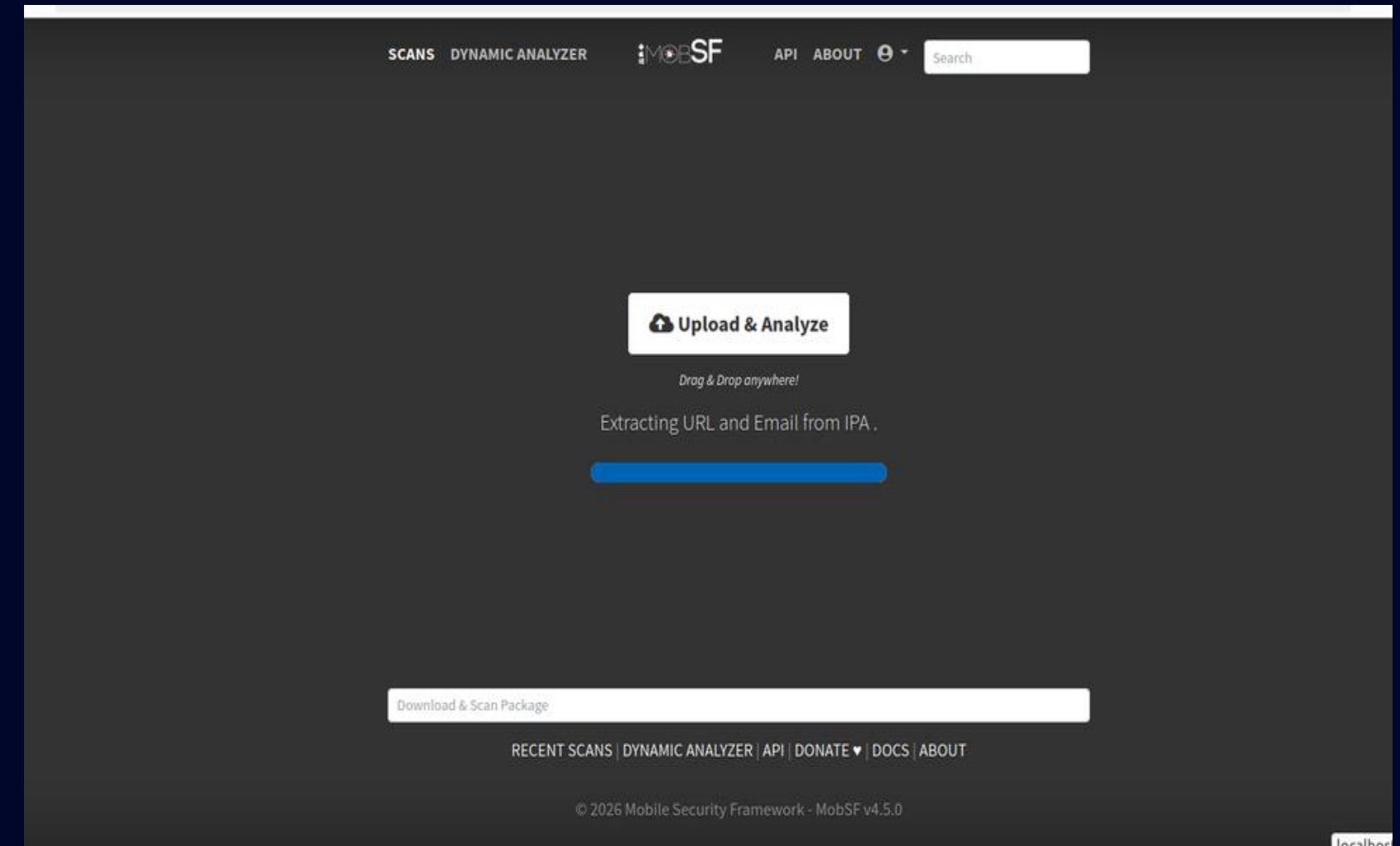
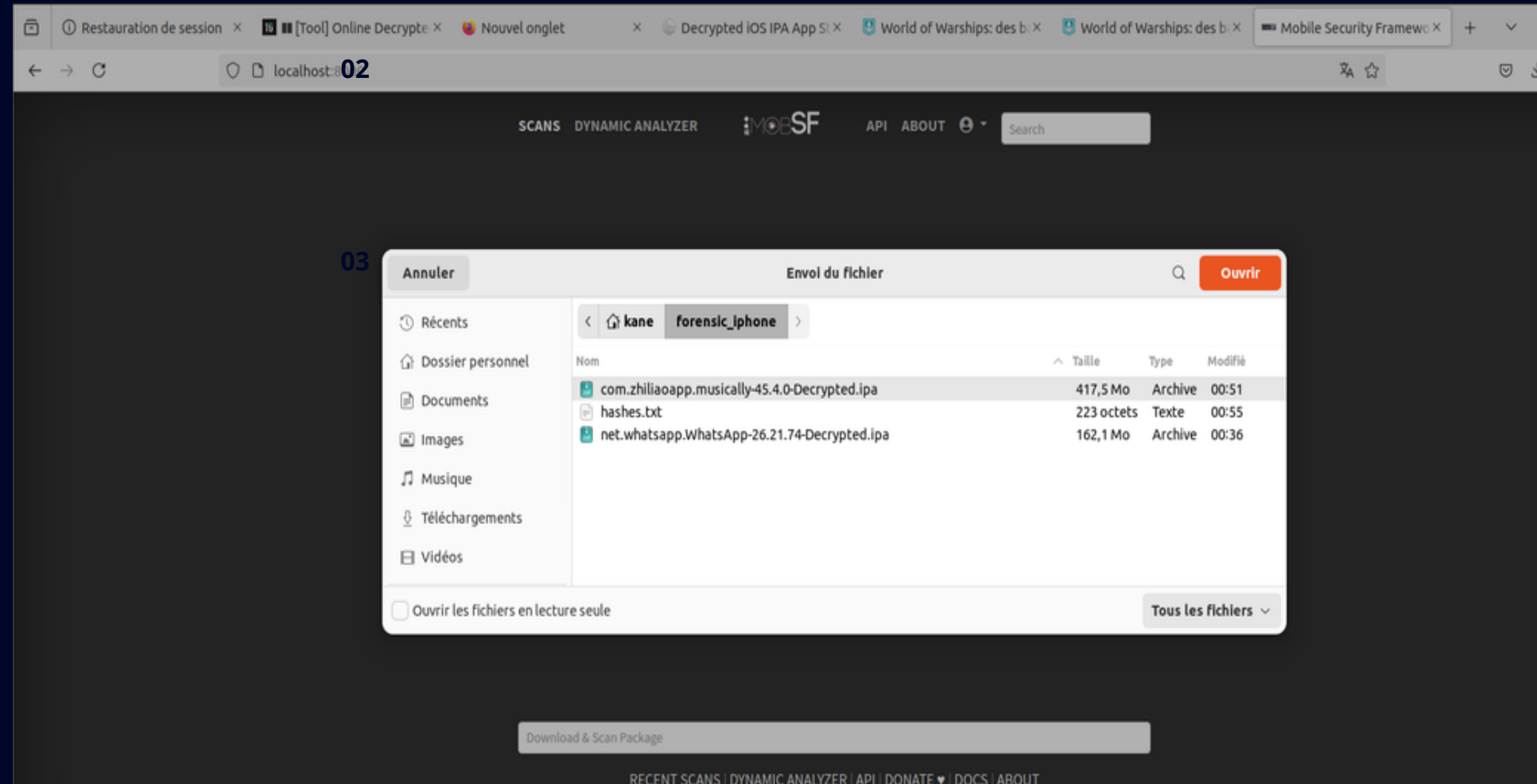
Upload via interface web
`http://localhost:8000` → Upload & Analyze → sélectionner le .ipa



Ou via API REST

`curl -F "file=@net.whatsapp.WhatsApp-26.21.74-Decrypted.ipa" \`
`http://localhost:8000/api/v1/upload \`
`-H "Authorization:`
`d5e9ade41005ac81bc179e04756e450231548d2af3bb90546638`
`7e5b8289fe55"`

Aller dans le plate forme mobsf puis Aller sur dossier personnel et choisir le fichier a analyser et commencer l'analyse





5 — Collecte des résultats

Static Analyzer

Information

Scan Options

URL Schemes

Permissions

Security Analysis

Malware Analysis

Reconnaissance

Libraries

Files

PDF Report

Print Report

APP SCORES

Security Score: 52/100

Trackers Detection: 0/432

MobSF Scorecard

FILE INFORMATION

File Name

net.whatsapp.WhatsApp-26.21.74-Decrypted.ipa

Size

154.64MB

Md5

99d54b5d89f4646b827faa10883a50ca

SHA1

1fab4240bd5942e31feb9d39830104ffc69e34c

SHA256

ee88f31441dcc10bb4fk205fb587bf83d872afc77762444d67865a5031d96ed0

APP INFORMATION

App Name

WhatsApp

App Type

Swift

Identifier

net.whatsapp.WhatsApp

SDK Name

iphonios26.2

Version

26.21.74

Build

985245296

Platform Version

26.2

Min OS Version

15.1

Supported Platforms

iPhoneOS,

BINARY INFORMATION

Arch

ARM64

SubArch

CPU_SUBTYPE_ARM64_ALL

Bit

64-bit

Endian

<

APP STORE INFORMATION

Title

WhatsApp Messenger

Score

4,68724

Features

iosUniversal,

Price

0.0

Category

Social Networking, Utilities,

App Store URL

net.whatsapp.WhatsApp

Developer

WhatsApp Inc.,

Developer ID

310634000

Developer Website

http://www.whatsapp.com/

Developer URL

https://apps.apple.com/us/developer/whatsapp-inc/id310634000?uo=4

Supported Devices

MacDesktop-MacDesktop, iPhone5s-iPhone5s, iPadAir-iPadAir, iPadAirCellular-iPadAirCellular, iPadMiniRetina-iPadMiniRetina, iPadMiniRetinaCellular-iPadMiniRetinaCellular, iPhone6-iPhone6, iPhone6Plus-iPhone6Plus, iPadAir2-iPadAir2, iPadAir2Cellular-iPadAir2Cellular, iPadMini3-iPadMini3, iPadMini3Cellular-iPadMini3Cellular, iPodTouchSixthGen-iPodTouchSixthGen, iPhone6s-iPhone6s, iPhone6sPlus-iPhone6sPlus, iPadMini4-iPadMini4, iPadMini4Cellular-iPadMini4Cellular, iPadPro-iPadPro, iPadProCellular-iPadProCellular, iPadPro97-iPadPro97, iPadPro97Cellular-iPadPro97Cellular, iPhoneSE-iPhoneSE, iPhone7-iPhone7, iPhone7Plus-iPhone7Plus, iPad611-iPad611, iPad612-iPad612, iPad71-iPad71, iPad72-iPad72, iPad73-iPad73, iPad74-iPad74, iPhone8-iPhone8, iPhone8Plus-iPhone8Plus, iPhoneX-iPhoneX, iPad75-iPad75, iPad76-iPad76, iPhoneXS-iPhoneXS, iPhoneXSMax-iPhoneXSMax, iPhoneXR-iPhoneXR, iPad812-iPad812, iPad834-iPad834, iPad856-iPad856, iPad878-iPad878, Watch4-Watch4, iPadMini5-iPadMini5, iPadMini5Cellular-iPadMini5Cellular, iPadAir3-iPadAir3, iPadAir3Cellular-iPadAir3Cellular, iPodTouchSeventhGen-iPodTouchSeventhGen, iPhone11-iPhone11, iPhone11Pro-iPhone11Pro, iPadSeventhGen-iPadSeventhGen, iPadSeventhGenCellular-iPadSeventhGenCellular, iPhone11ProMax-iPhone11ProMax, iPhoneSESecondGen-iPhoneSESecondGen, iPadProSecondGen-iPadProSecondGen, iPadProSecondGenCellular-iPadProSecondGenCellular, iPadProFourthGen-iPadProFourthGen, iPadProFourthGenCellular-iPadProFourthGenCellular, iPhone12Mini-iPhone12Mini, iPhone12-iPhone12, iPhone12Pro-iPhone12Pro, iPhone12ProMax-iPhone12ProMax, iPadAir4-iPadAir4, iPadAir4Cellular-iPadAir4Cellular, iPadElghthGen-iPadElghthGen, iPadElghthGenCellular-iPadElghthGenCellular, iPadProThirdGen-iPadProThirdGen, iPadProThirdGenCellular-iPadProThirdGenCellular, iPadProFifthGen-iPadProFifthGen, iPadProFifthGenCellular-iPadProFifthGenCellular, iPhone13Pro-iPhone13Pro, iPhone13ProMax-iPhone13ProMax, iPhone13Mini-iPhone13Mini, iPhone13-iPhone13, iPadMiniSixthGen-iPadMiniSixthGen, iPadNinthGen-iPadNinthGen, iPadNinthGenCellular-iPadNinthGenCellular, iPhoneSEThirdGen-iPhoneSEThirdGen, iPadAirFifthGen-iPadAirFifthGen, iPadAirFifthGenCellular-iPadAirFifthGenCellular, iPhone14-iPhone14, iPhone14Plus-iPhone14Plus, iPhone14Pro-iPhone14Pro, iPhone14ProMax-iPhone14ProMax, iPadTenthGen-iPadTenthGen, iPadTenthGenCellular-iPadTenthGenCellular, iPadPro11FourthGen-iPadPro11FourthGen, iPadPro11FourthGenCellular-iPadPro11FourthGenCellular, iPadProSixthGen-iPadProSixthGen, iPadProSixthGenCellular-iPadProSixthGenCellular, iPhone15-iPhone15, iPhone15Plus-iPhone15Plus, iPhone15Pro-iPhone15Pro, iPhone15ProMax-iPhone15ProMax, iPadAir11M2-iPadAir11M2, iPadAir11M2Cellular-iPadAir11M2Cellular, iPadAir13M2-iPadAir13M2, iPadAir13M2Cellular-iPadAir13M2Cellular, iPadPro11M4-iPadPro11M4, iPadPro11M4Cellular-iPadPro11M4Cellular, iPadPro13M4-iPadPro13M4, iPadPro13M4Cellular-iPadPro13M4Cellular, iPhone16-iPhone16, iPhone16Plus-iPhone16Plus, iPhone16Pro-iPhone16Pro, iPhone16ProMax-iPhone16ProMax, iPadMiniA17Pro-iPadMiniA17Pro, iPadMiniA17ProCellular-iPadMiniA17ProCellular, iPhone16e-iPhone16e, iPadA16-iPadA16, iPadA16Cellular-iPadA16Cellular, iPadAir11M3-iPadAir11M3, iPadAir11M3Cellular-iPadAir11M3Cellular, iPadAir13M3-iPadAir13M3, iPadAir13M3Cellular-iPadAir13M3Cellular, iPhone17Pro-iPhone17Pro, iPhone17ProMax-iPhone17ProMax, iPhone17-iPhone17, iPhoneAir-iPhoneAir, iPadPro11M5-iPadPro11M5, iPadPro11M5Cellular-iPadPro11M5Cellular, iPadPro13M5-iPadPro13M5, iPadPro13M5Cellular-iPadPro13M5Cellular, iPhone17e-iPhone17e, iPadAir11M4-iPadAir11M4, iPadAir11M4Cellular-iPadAir11M4Cellular, iPadAir13M4-iPadAir13M4, iPadAir13M4Cellular-iPadAir13M4Cellular,

Description

WhatsApp from Meta is a free messaging and calling app used by over 2 billion people across 180+ countries. It's super simple, reliable and private, making it perfect for staying in touch with your friends and family worldwide.

Transfert des données depuis localhost...

MobSF

RECENT SCANS

STATIC ANALYZER

DYNAMIC ANALYZER

API

DONATE

DOCS

ABOUT

Search

Static Analyzer

Information

Scan Options

URL Schemes

Permissions

Security Analysis

Malware Analysis

Reconnaissance

Libraries

Files

PDF Report

Print Report

WhatsApp from Meta is a free messaging and calling app used by over 2 billion people across 180+ countries. It's super simple, reliable and private, making it perfect for staying in touch with your friends and family worldwide. WhatsApp works across mobile, tablet, and desktop even on slow connections, with no subscription fees*.

Private messaging and calls across the world

Your privacy is our priority. With end-to-end encryption, you can be sure that your personal messages and calls stay between you and who you send them to. And no one, not even WhatsApp, can read or listen to them.

Simple and secure connections, right away

All you need is your phone number - no usernames or logins needed. Easily view your contacts on WhatsApp and start messaging. And what's more, you can easily link your other devices, including iPads, for seamless connectivity.

High quality voice and video calls

Enjoy free, secure video and voice calls with up to 32 people for free. Your calls work across devices, even on slower connections, using your phone or tablet's Internet service.

Group chats to keep you in contact

Stay close to friends and family through end-to-end encrypted group chats and calls. Share messages, photos, videos and documents across mobile, tablet, and desktop. You can also use call links and screen-sharing to make group calls and collaboration easy.

Stay connected in real time

You can share your location with those in your individual or group chats while using WhatsApp on your mobile phone. Or record a voice message for those moments when just text isn't good enough.

Share daily moments through Status

Share your everyday life with those who matter. With Status you can share text, photos, video, and GIF updates that disappear after 24 hours. You're always in control of who sees your status. Just choose who you want to share it with - all your contacts or selected ones. And bring your everyday to life.

*Data charges may apply. Contact your provider for details.

If you have any feedback or questions, please go to WhatsApp > Settings > Help > Contact Us

Terms of Service: <https://www.whatsapp.com/legal/terms-of-service>

Learn more about messaging privately: <https://www.whatsapp.com/privacy>

Learn more about WhatsApp safety: <https://www.whatsapp.com/security>

Transfert des données depuis localhost...

MobSF

RECENT SCANS

STATIC ANALYZER

DYNAMIC ANALYZER

API

DONATE

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ABOUT

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PDF Report

Print Report

SCAN OPTIONS

Rescan

Manage Suppressions

Scan Logs

DECOMPILED ASSETS

View Info.plist

View Class Dump

Download IPA

CUSTOM URL SCHEMES

URL NAME	SCHEMES
net.whatsapp.WhatsApp	whatsapp-consumer
Editor	upl
	whatsapp
	fb306069495113
	im

APPLICATION PERMISSIONS

PERMISSIONS	STATUS	INFO	REASON IN MANIFEST
NSBluetoothPeripheralUsageDescription	dangerous	Access Bluetooth Interface.	This lets you connect to supported Bluetooth devices
NSCalendarsUsageDescription	dangerous	Access Calendars.	This lets you create events on your calendar.
NSCameraUsageDescription	dangerous	Access the Camera.	This lets you make video calls, take photos, and record videos.

localhost:8000/static_analyzer_ios/99d54b5d89f4646b827faa10883a50ca/#files



5 — Collecte des résultats

APPLICATION PERMISSIONS			
PERMISSIONS	STATUS	INFO	REASON IN MANIFEST
NSBluetoothPeripheralUsageDescription	dangerous	Access Bluetooth Interface.	This lets you connect to supported Bluetooth devices
NSCalendarsUsageDescription	dangerous	Access Calendars.	This lets you create events on your calendar.
NSCameraUsageDescription	dangerous	Access the Camera.	This lets you make video calls, take photos, and record videos.
NSContactsUsageDescription	dangerous	Access Contacts.	Upload your contacts to WhatsApp's servers to help you quickly get in touch with your friends and help us provide a better experience.
NSFaceIDUsageDescription	normal	Access the ability to authenticate with Face ID.	Use Face ID to authenticate on WhatsApp.
NSLocationAlwaysUsageDescription	dangerous	Access location information at all times.	If you always allow access to your location, you can choose to share your live location, and it will update even when you're not using the app. This also lets you see your current location or a nearby place.
NSLocationUsageDescription	dangerous	Access location information at all times (< iOS 8).	This lets you send your current location or nearby places in chats.
NSLocationWhenInUseUsageDescription	dangerous	Access location information when app is in the foreground.	This lets you send your current location or nearby places in chats.
NSMicrophoneUsageDescription	dangerous	Access microphone.	This lets you make calls, send voice messages, and record videos with sound.
NSPhotoLibraryUsageDescription	dangerous	Access the user's photo library.	This lets you send photos and videos from your library and save the ones you capture.
NSRemindersUsageDescription	dangerous	Access the user's reminders.	This lets you set reminders to call your friends back.
NSSiriUsageDescription	dangerous	Allow app to send user data to Siri	This lets you use Siri to quickly send and read messages and make calls.
NSSpeechRecognitionUsageDescription	normal	Allow app to send user data to Apple's speech recognition servers.	WhatsApp does on-device recognition for Voice Messages. In this mode, no voice data is sent to Apple despite this alert, which is unexpected. Select "Don't Allow" in the alert.

Transfert des données depuis...

ie pointer inside or press Ctrl+G.

Static Analyzer

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Print Report

APP TRANSPORT SECURITY (ATS)

HIGH1

WARNING0

INFO0

SECURE0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App Transport Security AllowsArbitraryLoads is allowed	high	App Transport Security restrictions are disabled for all network connections. Disabling ATS means that unsecured HTTP connections are allowed. HTTPS connections are also allowed, and are still subject to default server trust evaluation. However, extended security checks like requiring a minimum Transport Layer Security (TLS) protocol version—are disabled. This setting is not applicable to domains listed in NSExceptionDomains.

IPA BINARY CODE ANALYSIS

HIGH0

WARNING3

INFO1

SECURE0

SUPPRESSED0

NO	ISSUE	SEVERITY	STANDARDS	DESCRIPTION	OPTIONS
1	Binary makes use of insecure API(s)	warning	CWE: CWE-676: Use of Potentially Dangerous Function OWASP Top 10: M7: Client Code Quality OWASP MASVS: MSTG-CODE-8	The binary may contain the following insecure API(s) _chmod , _fopen , _memcpy , _printf , _sprintf , _scanf , _stat , _strcat , _strcpy , _strlen , _strncpy , _vsprintf	
2	Binary makes use of the insecure Random function(s)	warning	CWE: CWE-330: Use of Insufficiently Random Values OWASP Top 10: M5: Insufficient Cryptography	The binary may use the following insecure Random function(s) _srand	

Log File			
OWASP MASVS: MSTG-STORAGE-3			
4	Binary makes use of malloc function	warning	CWE: CWE-789: Uncontrolled Memory Allocation OWASP Top 10: M7: Client Code Quality OWASP MASVS: MSTG-CODE-8

IPA BINARY ANALYSIS			
PROTECTION	STATUS	SEVERITY	DESCRIPTION
NX	True	info	The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.
PIE	True	info	The binary is build with -fPIC flag which enables Position Independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.
STACK CANARY	True	info	This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.
ARC	True	info	The binary is compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.
RPATH	True	warning	The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.
CODE SIGNATURE	True	info	This binary has a code signature.
ENCRYPTED	False	warning	This binary is not encrypted.
SYMBOLS STRIPPED	False	warning	Debug Symbols are available. To strip debugging symbols, set Strip Debug Symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.

DYNAMIC LIBRARY & FRAMEWORK BINARY ANALYSIS			
Transfert des données depuis localhost...			

Static Analyzer			
Transfert des données depuis localhost...			

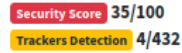
NO	DYLIB/Framework	NX	STACK CANARY	ARC	RPATH	CODE SIGNATURE	ENCRYPTED	SYMBOLS STRIPPED
1	SharedModules.framework/SharedModules	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	True info The binary is compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.	True warning The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.	True info This binary has a code signature.	False warning This binary is not encrypted.	False warning Debug Symbols are available. To strip debugging symbols, set Strip Debug Symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.
2	PspiphonTunnel.framework/PspiphonTunnel	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	True info The binary is compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.	True warning The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.	True info This binary has a code signature.	False warning This binary is not encrypted.	False warning Debug Symbols are available. To strip debugging symbols, set Strip Debug Symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.

FILE ANALYSIS		
NO	ISSUE	FILES
1	Plist Files	Show Files

FIREBASE DATABASE ANALYSIS		
Transfert des données depuis localhost...		

SEVERITY		DESCRIPTION

FILE INFORMATION



File Name	com.zhiliaoapp.musically-45.4.0-Decrypted.ipa
Size	398.12MB
MD5	b829769649e179b082bb366e3260ce4
SHA1	1a335cad26e581787a30ccdaeb9592d7c25e5b0
SHA256	fc6db9b0ea67d65d1fd3bc95c2c95e02b8dd4ef750d26e2811597cacd3d69e2

iAPP INFORMATION

App Name	TikTok				
App Type	Swift				
Identifier	com.zhiliaoapp.musically				
SDK Name	iphones26.0				
Version	45.4.0	Build	454024	Platform Version	26.0
				Min OS Version	14.0
Supported Platforms	iPhoneOS,				

Ad BINARY INFORMATION

Arch ARM64
Sub Arch CPU_SUBTYPE_ARM64_ALL
Bit 64-bit Endian <

APP STORE INFORMATION

Title	TikTok - Videos, Shop & LIVE				
Score	4.72829	Features	ios,universal	Price	0.0
		Category	Entertainment, Photo & Video		
Developer	TikTok Ltd.,	Developer ID	1322881000		
Developer Website	http://www.tiktok.com				
Developer URL	https://apps.apple.com/us/developer/tiktok-ltd/id1322881000?uo=4				

Supported Devices: iPhone5s-iPhone5s, iPadAir-iPadAir, iPadAirCellular-iPadAirCellular, iPadMiniRetina-iPadMiniRetina, iPadMiniRetinaCellular-iPadMiniRetinaCellular, iPhone6-iPhone6, iPhone6Plus-iPhone6Plus, iPadAir2-iPadAir2, iPadAir2Cellular-iPadAir2Cellular, iPadMini3-iPadMini3, iPadMini3Cellular-iPadMini3Cellular, iPodTouchSixthGen-iPodTouchSixthGen, iPhone6s-iPhone6s, iPhone6sPlus-iPhone6sPlus, iPadMini4-iPadMini4, iPadMini4Cellular-iPadMini4Cellular, iPadPro-iPadPro, iPadProCellular-iPadProCellular, iPadPro97-iPadPro97, iPadPro97Cellular-iPadPro97Cellular, iPhoneSE-iPhoneSE, iPhone7-iPhone7, iPhone7Plus-iPhone7Plus, iPad611-iPad611, iPad612-iPad612, iPad71-iPad71, iPad72-iPad72, iPad73-iPad73, iPad74-iPad74, iPhone8-iPhone8, iPhone8Plus-iPhone8Plus, iPhoneX-iPhoneX, iPad75-iPad75, iPad76-iPad76, iPhoneXS-iPhoneXS, iPhoneXSMax-iPhoneXSMax, iPhoneXR-iPhoneXR, iPad812-iPad812, iPad834-iPad834, iPad856-iPad856, iPad878-iPad878, iPadMini5-iPadMini5, iPadMini5Cellular-iPadMini5Cellular, iPadAir3-iPadAir3, iPadAir3Cellular-iPadAir3Cellular, iPodTouchSeventhGen-iPodTouchSeventhGen, iPhone11-iPhone11, iPhone11Pro-iPhone11Pro, iPadSeventhGen-iPadSeventhGen, iPadSeventhGenCellular-iPadSeventhGenCellular, iPhone11ProMax-iPhone11ProMax, iPhoneSESecondGen-iPhoneSESecondGen, iPadProSecondGen-iPadProSecondGen, iPadProSecondGenCellular-iPadProSecondGenCellular, iPadProFourthGen-iPadProFourthGen, iPadProFourthGenCellular-iPadProFourthGenCellular, iPhone12Mini-iPhone12Mini, iPhone12-iPhone12, iPhone12Pro-iPhone12Pro, iPhone12ProMax-iPhone12ProMax, iPadAir4-iPadAir4, iPadAir4Cellular-iPadAir4Cellular, iPadEightGen-iPadEightGen, iPadEightGenCellular-iPadEightGenCellular, iPadProThirdGen-iPadProThirdGen, iPadProThirdGenCellular-iPadProThirdGenCellular, iPadProFifthGen-iPadProFifthGen, iPadProFifthGenCellular-iPadProFifthGenCellular, iPhone13Pro-iPhone13Pro, iPhone13ProMax-iPhone13ProMax, iPhone13Mini-iPhone13Mini, iPhone13-iPhone13, iPadMiniSixthGen-iPadMiniSixthGen, iPadMiniSixthGenCellular-iPadMiniSixthGenCellular, iPadNinthGen-iPadNinthGen, iPadNinthGenCellular-iPadNinthGenCellular, iPhoneSEThirdGen-iPhoneSEThirdGen, iPadAirFifthGen-iPadAirFifthGen, iPadAirFifthGenCellular-iPadAirFifthGenCellular, iPad14-iPhone14, iPhone14Plus-iPhone14Plus, iPhone14Pro-iPhone14Pro, iPhone14ProMax-iPhone14ProMax, iPadTenthGen-iPadTenthGen, iPadTenthGenCellular-iPadTenthGenCellular, iPadPro11FourthGen-iPadPro11FourthGenCellular, iPadPro11FourthGenCellular, iPadProSixthGen-iPadProSixthGen, iPadProSixthGenCellular-iPadProSixthGenCellular, iPhone15-iPhone15, iPhone15Plus-iPhone15Plus, iPhone15Pro-iPhone15Pro, iPhone15ProMax-iPhone15ProMax, iPadAir11M2-iPadAir11M2, iPadAir11M2Cellular-iPadAir11M2Cellular, iPadAir13M2-iPadAir13M2, iPadAir13M2Cellular-iPadAir13M2Cellular, iPadPro11M4-iPadPro11M4, iPadPro11M4Cellular-iPadPro11M4Cellular, iPadPro13M4-iPadPro13M4, iPadPro13M4Cellular-iPadPro13M4Cellular, iPhone16-iPhone16, iPhone16Plus-iPhone16Plus, iPhone16Pro-iPhone16Pro, iPhone16ProMax-iPhone16ProMax, iPadMiniA17Pro-iPadMiniA17Pro, iPadMiniA17ProCellular-iPadMiniA17ProCellular, iPhone16e-iPhone16e, iPadA16-iPadA16, iPadA16Cellular-iPadA16Cellular, iPadAir11M3-iPadAir11M3, iPadAir11M3Cellular-iPadAir11M3Cellular, iPadAir13M3-iPadAir13M3, iPadAir13M3Cellular-iPadAir13M3Cellular, iPhone17Pro-iPhone17Pro, iPhone17ProMax-iPhone17ProMax, iPhone17-iPhone17, iPhoneAir-iPhoneAir, iPadPro11M5-iPadPro11M5, iPadPro11M5Cellular-iPadPro11M5Cellular, iPadPro13M5-iPadPro13M5, iPadPro13M5Cellular-iPadPro13M5Cellular, iPhone17e-iPhone17e, iPadAir11M4-iPadAir11M4, iPadAir11M4Cellular-iPadAir11M4Cellular, iPadAir13M4-iPadAir13M4, iPadAir13M4Cellular-iPadAir13M4Cellular,

TikTok is the global discovery platform for videos, streaming on LIVE, shopping, and more. Our mission is to inspire creativity and bring joy. Over a billion people come to TikTok to share what they love, learn new things, and find ideas that make everyday life more interesting.

SCAN OPTIONS

Rescan

Manage Suppressions

Scan Logs

DECOMPILED ASSETS

View Info.plist

View Class Dump

Download IPA

CUSTOM URL SCHEMES

URL NAME	SCHEMES
sso Editor	lkz65bruc1383kpguy
Helo Editor	helodad736bdd1af7bd7e3edfc29eb8d6e11
com.zhiliaapp.musically Editor	snssdk1233
com.zhiliaapp.musically Editor	snssdkonly1233
com.zhiliaapp.musically.usm Editor	snssdkm21233
com.zhiliaapp.musically.liteusm Editor	snssdklite21233
com.zhiliaapp.musically.otl Editor	snssdkotl1233
facebook Editor	fb597615686992125 ak597615686992125
line Editor	line3rdp.com.zhiliaapp.musically

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APPLICATION PERMISSIONS

PERMISSIONS	STATUS	INFO	REASON IN MANIFEST
NSAppleMusicUsageDescription	dangerous	Access Apple Media Library.	To select sound from your library, allow us to access your Apple Music.
NSBluetoothPeripheralUsageDescription	dangerous	Access Bluetooth Interface.	This will allow "TikTok" to find and connect to Bluetooth accessories. This app may also use Bluetooth to connect with nearby devices.
NSCalendarsUsageDescription	dangerous	Access Calendars.	Add events to your device calendar to get reminders when events start.
NSCameraUsageDescription	dangerous	Access the Camera.	You'll be able to record video.
NSContactsUsageDescription	dangerous	Access Contacts.	Sync your contacts to easily find people you know on TikTok. Your contacts will only be used to help you connect with friends.
NSFaceIDUsageDescription	normal	Access the ability to authenticate with Face ID.	We'll use this to unlock features that require added security.
NSLocationWhenInUseUsageDescription	dangerous	Access location information when app is in the foreground.	If location services are enabled, TikTok will collect, store and use your device's approximate location to improve your experience.
NSMicrophoneUsageDescription	dangerous	Access microphone.	To record audio, allow us to access your Microphone.
NSPhotoLibraryUsageDescription	dangerous	Access the user's photo library.	To upload from or download to your device, allow access to your photos in your phone settings.
NSLocalNetworkUsageDescription	normal	Allow app to request access to the local network.	The TikTok app will be able to discover and connect with devices that have the TikTok app installed for camera and screen sharing.

APP TRANSPORT SECURITY (ATS)

HIGH2

WARNING1

INFO1

SECURE0

NO	ISSUE	SEVERITY	DESCRIPTION
1	Insecure WebView load is allowed	high	App Transport Security restrictions are disabled for requests made from WebViews without affecting your URLSession connections. This setting is not applicable to domains listed in NSExceptionDomains.
2	NSExceptionDomains	info	zero-rating.tiktok.com

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2	NSExceptionDomains	info	zero-rating.tiktok.com
3	Insecure communication to zero-rating.tiktok.com is allowed	high	NSExceptionAllowsInsecureHTTPLoads allows insecure HTTP loads to zero-rating.tiktok.com, or to be able to loosen the server trust evaluation requirements for HTTPS connections to the domain.
4	NSRequiresCertificateTransparency set to NO for zero-rating.tiktok.com	warning	Certificate Transparency (CT) is a protocol that ATS can use to identify mistakenly or maliciously issued X.509 certificates. Set the value for the NSRequiresCertificateTransparency key to YES to require that for a given domain, server certificates are supported by valid, signed CT timestamps from at least two CT logs trusted by Apple. This key is optional. The default value is NO.

IPA BINARY CODE ANALYSIS

NO	ISSUE	SEVERITY	STANDARDS	DESCRIPTION	OPTIONS
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IPA BINARY ANALYSIS

PROTECTION	STATUS	SEVERITY	DESCRIPTION
NX	True	info	The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.
PIE	True	info	The binary is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.
STACK CANARY	False	warning	This binary has debug symbols stripped. We cannot identify whether stack canary is enabled or not.
ARC	True	info	The binary is compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.
RPATH	True	warning	The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.
CODE SIGNATURE	True	info	This binary has a code signature.
ENCRYPTED	False	warning	This binary is not encrypted.

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		memory page non-executable making attacker injected shellcode non-executable.	overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.		signature.	encrypted.	Debug symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.
1	BDLRepaier.framework/BDLRepaier	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	False warning This binary has debug symbols stripped. We cannot identify whether stack canary is enabled or not.	False warning This binary has debug symbols stripped. We cannot identify whether ARC is enabled or not.	True warning The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.	True info This binary has a code signature.	False warning This binary is not encrypted.	True info Debug Symbols are stripped
2	WCDBSQLCipherFrameworkA.framework/WCDBSQLCipherFrameworkA	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	False high The binary is not compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and protects from memory corruption vulnerabilities. Use compiler option -fobjc-arc to enable ARC or set Objective-C Automatic Reference Counting to YES in project configuration.	True warning The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.	True info This binary has a code signature.	False warning This binary is not encrypted.	False warning Debug Symbols are available. To strip debugging symbols, set Strip Debug Symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.
3	OMSDK_Bytedance1.framework/OMSDK_Bytedance1	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	True info The binary is compiled with Automatic Reference Counting (ARC) flag. ARC is a compiler feature that provides automatic memory management of Objective-C objects and is an exploit mitigation mechanism against memory corruption vulnerabilities.	True warning The binary has Runpath Search Path (@rpath) set. In certain cases an attacker can abuse this feature to run arbitrary executable for code execution and privilege escalation. Remove the compiler option -rpath to remove @rpath.	True info This binary has a code signature.	False warning This binary is not encrypted.	False warning Debug Symbols are available. To strip debugging symbols, set Strip Debug Symbols During Copy to YES, Deployment Postprocessing to YES, and Strip Linked Product to YES in project's build settings.
4	AAWELaunchTracker.framework/AAWELaunchTracker	True info	False high	False high	True warning	True info	False warning	False warning



L'analyse forensique des applications WhatsApp et TikTok installées sur un iPhone 14 Plus sous iOS 18.7.8 nous a permis de mettre en évidence l'importance de la sécurité des applications mobiles dans notre quotidien numérique

Ce que nous avons appris

- Les applications mobiles collectent de nombreuses données personnelles via leurs permissions
- Même des applications populaires et largement utilisées peuvent présenter des vulnérabilités
- L'analyse forensique mobile nécessite une méthodologie rigoureuse et des outils spécialisés
- La chaîne de custody est essentielle pour garantir la validité des preuves numériques